

Efficient Speech Watermarking for Speech Synthesis via Progressive Knowledge Distillation

Yang Cui, Peter Pan, Lei He, Sheng Zhao

Microsoft, Voice AI



PKDMark: A compact neural watermarking model achieving high level robustness (F1 score 99.6%) and imperceptibility (PESQ 4.30) with a 93.6% reduction in computational cost.

INTRODUCTION

Unauthorized voice cloning poses significant privacy and security risks

As speech generative models (TTS, voice cloning) become increasingly realistic, distinguishing natural and synthetic speech is becoming difficult.

This raises concerns around:

- Privacy security
- Copyright protection
- Deepfake misinformation
- Unauthorized content reproduction

Speech watermarking offers a viable solution

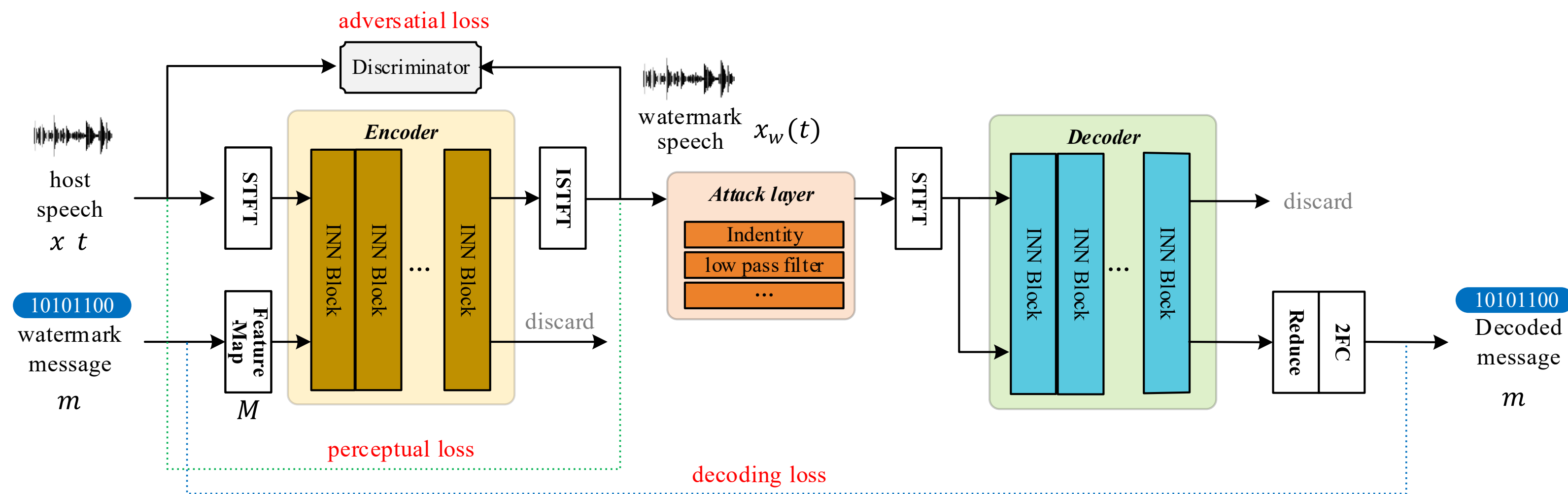
Speech watermarking embeds imperceptible, traceable information into audio, enabling **source tracing** and **misuse prevention**.

PROBLEM: Efficiency Gap

Audio watermarking generally includes DSP-based and neural network-based methods, each characterized by its own advantages and constraints.

	DSP-based methods	Neural-based methods
Pros	Computationally efficient for real-time processing	Achieve strong balance of perceptual quality, robustness, and capacity
Cons	Often fragile under advanced distortions such as reverberation and frequency/time manipulations	Typically require large model size and high computational cost

METHOD



Encoder-Attack-Decoder Pipeline

- Encoder**: Fuses the host speech STFT and a complex-domain message feature map to produce a watermarked spectrogram.
- Attack Layer**: Simulates 14 real-world distortions (noise, MP3, low-pass, resampling, reverb, pitch/time modifications) to enforce robustness.
- Decoder**: Shares parameters with the encoder through the INN structure and reconstructs the embedded bits from the distorted watermarked audio.
- Discriminator**: Supplies adversarial feedback to encourage natural and imperceptible watermarked speech.

Multi-objective loss

- Perceptual loss**: Minimizes multi-scale STFT differences
- Decoding loss**: Ensures accurate and stable message recovery
- Discriminative loss**: Improves perceptual fidelity

PKDMark

Progressive Knowledge Distillation (PKD)

PKD enables the student model to **gradually** inherit the teacher's robustness and perceptual quality while **significantly** reducing computation.

- Progressive Training Strategy**: During training, the watermarked outputs of the teacher and student are linearly mixed with an increasing factor λ .

$$x_{mix}(t) = \lambda(n)x_w^S(t) + (1 - \lambda(n))x_w^T(t)$$

- Early stage (small λ)**: Teacher dominates the supervision, providing stable gradients and preventing the student from falling into poor local minima.
- Middle to late stage ($\lambda \rightarrow 1$)**: The student progressively takes over, learning to embed robust watermarks independently.
- Balanced Optimization**: Direct knowledge distillation often sacrifices either perceptual quality or robustness, whereas our gradual transition strategy enables the student model to achieve a **superior balance** between imperceptibility and robustness.

Complex Message Feature Map

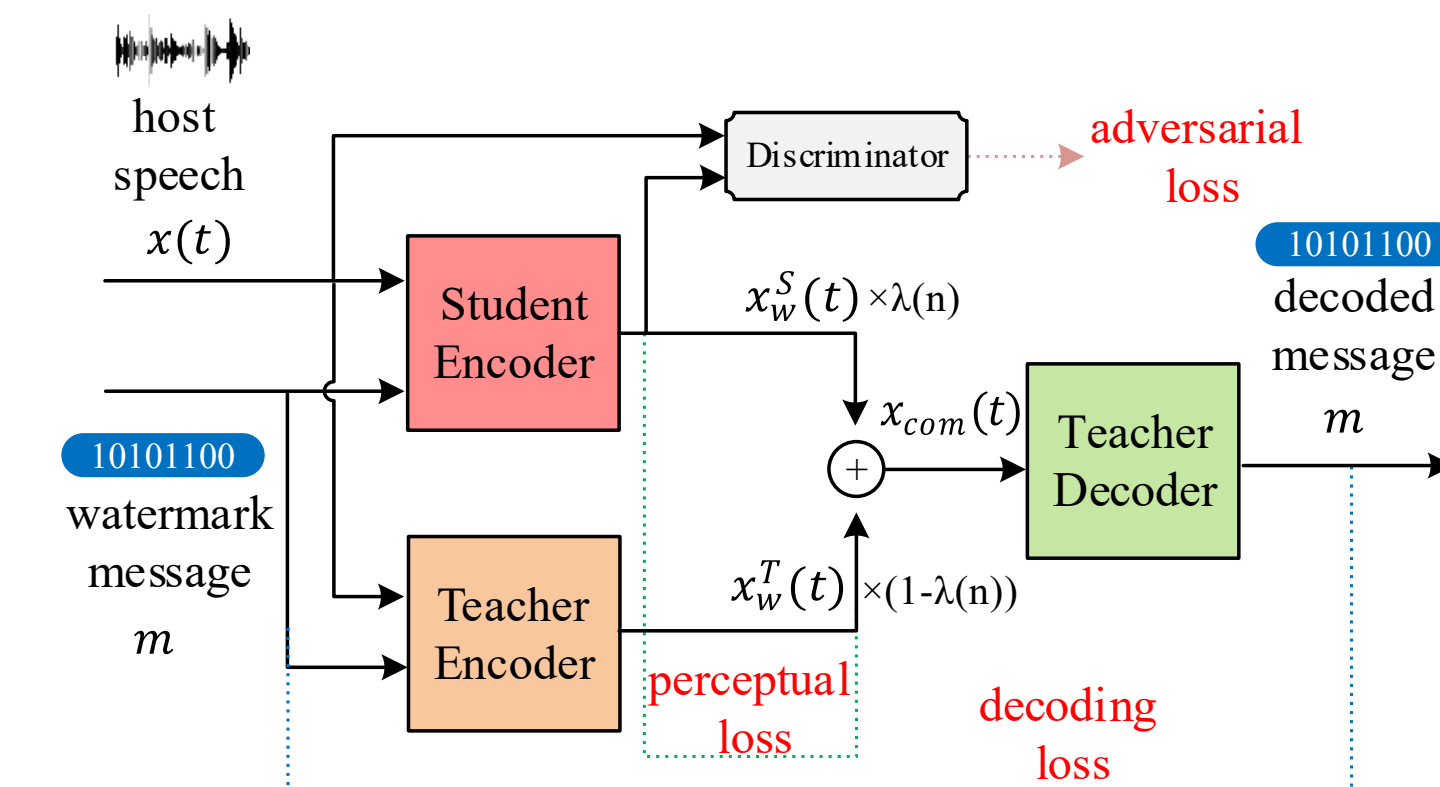
WavMark: Maps the message sequence to the time domain

Proposed: Directly embeds the message in the **complex STFT domain**, improving efficiency and aligning the feature map with the spectrogram time-frequency structure.

- Time-Frequency Expansion**: The message is expanded into a 2D feature map that matches the size of the host speech spectrogram.
- Complex-Domain Representation**: Watermark information is embedded in both the real and imaginary components, fully leveraging the complex STFT space.
 $M = Repeat_T(h + j \cdot h); j = \sqrt{-1}$
- Fine-Grained Alignment**: A dedicated message **feature map loss** enforces precise alignment between encoded and decoded message features, ensuring reliable message recovery.

Distortion types in attack layer

- Basic
 - ND — No Distortion
 - GN — Gaussian Noise (down to SNR 20dB)
 - MF — Median Filtering (5-point median filtering)
 - MP3 — MP3 Compression (down to 64 kbps)
 - LP — Low-Pass Filtering (4 kHz stopband)
 - DS — Downsampling (down to 8 kHz)
 - QT — Quantization (down to 9-bit)
 - AS — Amplitude Scaling (scaling factor $\in \{-1, +1\}$)
 - EA — Echo Addition (up to 100 ms delay)
 - SS — Slow Speed (down to 0.9x)
 - FS — Fast Speed (up to 1.1x)
- Advanced
 - RA — Reverberation Attack (up to RT60 = 200 ms)
 - PU — Pitch Up (up to +10%)
 - PD — Pitch Down (up to -10%)



RESULTS

Imperceptibility

Objective Measures:

- PESQ**: 4.30 (student) vs. 4.33 (teacher)
- SNR**: 45.54dB (student) vs. 45.75dB (teacher)

Subjective Measure:

- CMOS**: - 0.04 (Host speech vs PKDMark)

PESQ(↑): objective perceptual quality score

SNR(↑): signal-to-noise ratio to measure the watermark distortion level

CMOS: subjective comparison score indicating perceived quality difference

Score	Comparison
0	A is the same with B
-1 / +1	A is slightly worse / better than B
-2 / +2	A is worse / better than B
-3 / +3	A is much worse / better than B

Robustness

Models	BPS(↑)	PESQ(↑)	SNR(dB)(↑)	MEAN	ND	GN	MF	MP3	LP	DS	QT	AS	EA	SS	FS	RA	PU	PD
WavMark	32	4.31	39.99	8.58	0.19	1.17	3.24	0.69	1.07	1.92	1.96	0.32	0.21	2.70	3.69	5.56	49.46	47.94
AudioSeal	16	4.28	27.53	18.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.56	52.12	59.50	47.09	50.69	51.25
WavMark-16bps	16	4.29	41.56	2.43	0.54	1.14	2.52	0.84	0.78	1.26	1.32	0.54	0.60	3.25	3.06	9.56	5.23	6.49
Proposed teacher	16	4.33	45.75	0.56	0.00	0.38	0.46	0.13	0.07	0.85	0.13	0.00	0.00	1.11	1.20	1.45	1.04	0.98
PKDMark	16	4.30	45.54	0.51	0.00	0.39	0.58	0.19	0.32	1.23	0.00	0.00	0.00	0.52	0.85	1.58	0.65	0.84

Low BER Across 14 Distortions:

PKDMark achieves an **average BER of 0.51%**, comparable to the teacher model (0.56%) and significantly lower than WavMark and AudioSeal under advanced attacks.

Strong Performance Under Advanced Attacks:

Maintains stable decoding under time-stretching, pitch-shifting, reverberation, demonstrating strong generalization to real-world distortions.

Reliable Detection Accuracy:

Achieves an average **F1 score of 99.6%** across all 14 tested distortions, ensuring consistent and dependable watermark identification for speech synthesis.

Ablation study

Models	BPS(↑)	PESQ(↑)	SNR(dB)(↑)	MEAN	ND	GN	MF	MP3	LP	DS	QT	AS	EA	SS	FS	RA	PU	PD
Teacher (Time-Domain Feature Map)	16	4.30	42.25	1.77	0.60	0.84	1.68	0.84	0.78	0.96	0.90	0.54	1.02	1.86	1.86	7.03	3.43	4.45
Teacher (Complex-Domain Feature Map)	16	4.33	45.75	0.56	0.00	0.38	0.46	0.13	0.07	0.85	0.13	0.00	0.00	1.11	1.20	1.45	1.04	0.98
Student (Direct Knowledge Distillation)	16	4.08	44.29	1.94	0.60	2.41	1.50	0.91	2.93	6.38	0.39	0.07	0.20	3.13	3.19	2.93	4.10	2.54
Student (Progressive Knowledge Distillation)	16	4.30	45.54	0.51	0.00	0.39	0.58	0.19	0.32	1.23	0.00	0.00	0.00	0.52	0.85	1.58	0.65	0.84

Time-domain Feature Map vs Complex-domain Feature Map

- PESQ**: Complex-domain 4.33 vs Time-domain 4.30, comparable perceptual quality
- SNR**: Complex-domain achieves 45.75 dB > Time-domain 42.25 dB
- Robustness (BER)**: Average BER improves significantly from **1.77% → 0.56%**.
- Conclusion**: The complex-domain feature map provides substantially stronger robustness while preserving high perceptual quality.

Direct Knowledge Distillation vs Progressive Knowledge Distillation

- PESQ**: PKD 4.30 vs DKD 4.08, PKD perceptual quality is better than DKD.
- SNR**: PKD achieves 45.54 dB > DKD 44.29 dB
- Robustness (BER)**: Average BER improves from 1.94% (DKD) to 0.51% (PKD).
- Conclusion**: Progressive distillation delivers stronger robustness and better perceptual quality, outperforming direct distillation in both robustness and overall perceptual quality.

Computational Cost

- Lightweight**: PKDMark significantly reduces the computational burden of neural watermarking. Through progressive knowledge distillation, the lightweight student model achieves a 93.6% reduction compared to the teacher model (36.0 → 2.3 GFLOPs), while maintaining nearly identical perceptual quality and robustness.

- Real-Time Performance**: This efficiency gain enables real-time watermark embedding on both GPU (RTF 0.006 on Nvidia T4) and CPU (RTF 0.020 on Intel Core i9), making PKDMark practical for deployment in speech synthesis.

Model	GFLOPS	RTF (↓)	
		NVIDIA Tesla T4 (GPU)	Intel Core i9-13900
Teacher Model	36.0	0.039	0.155
PKDMark	2.3	0.006	0.020
Reduction (%)	93.6%	84.7%	87.1%